

zones ($15^\circ - 15^\circ$ meridian difference). Therefore, each time zone has a 1 hour time difference and each meridian has the difference of 4 minutes, moving each meridian east G.M.T. time is 4 minutes ahead and moving west of G.M.T. time remains 4 minutes behind for each meridian.

According to the question time is to be calculated for that place which is ahead ($6-3=3$ hrs) of Greenwich. As we know for 1 hours difference in longitude (meridian) is 15° therefore for 3 hours the difference will be $15 \times 3 = 45^\circ$, So that place will lie at 45° east of Greenwich.

24. What is the local time of Thimphu (Bhutan) located at 90° East longitude when the time at Greenwich (0°) is 12:00 noon :

- (a) 6:00 p.m. (b) 4.00 p.m.
(c) 7.00 p.m. (d) 6.00 a.m.

69th B.P.S.C. (Pre), 2023

Ans. (a)

Difference between Greenwich and Thimphu (Bhutan)
= 90° of longitudes
Total time difference = $90 \times 4 = 360$ minutes
 $\therefore$ The time increases at a rate of 4 minutes per one degree of longitude.
= $360 / 60$ hours = 6 hour
= 6 hours / Local time of Thimphu is 6 hours more than that at Greenwich i.e. 6.00 p.m.

25. Which one of the following is not a natural unit of time?

- (a) Tropical year (b) Lunar month
(c) Standard time (d) Day

Uttarakhand P.C.S. (Pre) 2016

Ans. (c)

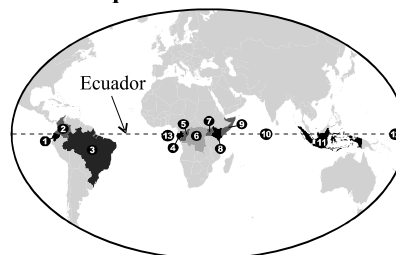
According to the question, the standard time has been artificially determined from the given alternatives. For example, the 0° longitude line has been determined as the prime meridian. Similarly, the $82\frac{1}{2}^\circ$ East longitude line has been set as the standard time of India, while the tropical year, the lunar month and the day are the natural units of time.

iii. The Equator

Latitudes are the angular distance of a point on the earth's surface, measured in degrees from the center of the earth. *An equator is the 0° circle of latitude (parallel). It is the imaginary line on the spheroid's surface, equidistant from the poles, dividing it into the northern and southern hemispheres. *On

earth, the equator is larger than any other latitude. Hence, it is a **great circle**. *As the latitudes move away from the equator towards the poles, their length decreases. The length of the earth's equator is appx. 40090 km (approx. 24911 miles). Out of its total length, 78.7% lies across water bodies and only 21.3% lies across land. The equatorial region receives almost vertical rays of the sun throughout the year, hence the difference between the maximum and minimum range of temperature is next to zero.

Equatorial Countries



- (1) Ecuador
- (2) Colombia
- (3) Brazil
- (4) Gabon
- (5) Democratic Republic of Congo
- (6) Republic of the Congo
- (7) Uganda
- (8) Kenya
- (9) Somalia
- (10) Maldives
- (11) Indonesia
- (12) Kiribati
- (13) Sao Tome and Principe

Tropic of Cancer, Equator and Tropic of Capricorn, all the three pass through Africa. The Tropic of Capricorn & Equator pass through South America.

1. The total length of Earth's equator is about –

- (a) 6,400 km (b) 12,800 km
(c) 40,000 km (d) 5,000 Em

Uttarakhand P.C.S. (Pre) 2010

Ans. (c)

The equator usually refers to an imaginary line, on the Earth's surface at equal distance from the North Pole and the South Pole, dividing the Earth into the Northern Hemisphere and Southern Hemisphere. The total length of the equator is 40090 km (approximately 24,911 miles). 78.7% of the line passes through water while 21.3% passes through land. Therefore, option (c) can be taken as the correct answer.

2. Through which, one of the following groups of countries does the Equator pass?

- (a) Brazil, Zambia and Malaysia
- (b) Colombia, Kenya and Indonesia
- (c) Brazil, Sudan and Malaysia
- (d) Venezuela, Ethiopia and Indonesia

I.A.S. (Pre) 2006

Ans. (b)

The Equator passes through 13 countries which are as follows:

- (1) Brazil
- (2) Colombia
- (3) Democratic Republic of the Congo
- (4) Ecuador
- (5) Gabon
- (6) Indonesia
- (7) Kenya
- (8) Kiribati
- (9) Republic of the Congo
- (10) Sao Tome and Principe
- (11) Somalia
- (12) Uganda
- (13) Maldives

From the above list, it is now clear that the Equator passes through Colombia, Kenya and Indonesia. Therefore, option (b) is the correct answer.

3. The equator does not pass through which of the following countries?

- (a) Kenya
- (b) Mexico
- (c) Indonesia
- (d) Brazil

Uttarakhand P.C.S. (Pre) 2021

Ans. (b)

See the explanation of the above question.

4. The equator does NOT pass through which of the following Countries?

- 1. Gabon
- 2. Somalia
- 3. Equatorial Guinea
- 4. Rwanda

Select the correct answer using the codes given below:

Codes :

- (a) 1 and 4
- (b) 2 and 4
- (c) 2 and 3
- (d) 3 and 4

U.P. P.C.S. (Pre) 2020

Ans. (d)

See the explanation of the above question.

5. The Equator does not pass through which of the following countries?

- (a) Kenya
- (b) Mexico
- (c) Indonesia
- (d) Brazil

M.P.P.C.S. (Pre) 2012

Ans. (b)

See the explanation of the above question.

6. Which of the following countries are located on the Equator?

- 1. Brunei
- 2. Colombia
- 3. Kenya
- 4. Venezuela

Select the correct answer using the codes given below:

Code :

- (a) 1 and 2
- (b) 2 and 3
- (c) 3 and 4
- (d) 1 and 4

U.P.P.C.S (Pre) 2010

Ans. (b)

See the explanation of the above question.

7. The equator passes through –

- (a) Cameroon
- (b) Costa Rica
- (c) Kenya
- (d) Venezuela

U.P.P.C.S. (Pre) 2008

U.P.P.C.S. (Mains) 2007

Ans. (c)

See the explanation of the above question.

8. The latitude at which 'Annual Range of Temperature remains minimum is –

- (a) Equator
- (b) Tropic of Cancer
- (c) Tropic of Capricorn
- (d) North Pole Circle

R.A.S./R.T.S. (Pre) 1999

Ans. (a)

Annual temperature difference at the Equator is minimum throughout the year. Since sun rays fall almost perpendicular throughout the year, there is little temperature fluctuation throughout the year. There is no winter season in the equatorial region.

9. Which one of the following cities is nearest to the equator?

- (a) Colombo
- (b) Jakarta
- (c) Manila
- (d) Singapore

I.A.S. (Pre) 2008

Ans. (d)

Colombo is situated at 6°55'30" North latitude, Jakarta at 6°12' South latitude, Manila at 14°35'30" North latitude and Singapore lies at 1°21' North latitude. Therefore, Singapore lies closest to the Equator (0° latitude). Singapore is situated only at a distance of 137 km from the Equator.